Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
1	*(a)*	(i)	Quarks **and** leptons **and** mesons – all required	1			1		
		(ii)	Quarks **and** mesons	1			1		
	(b)	(i)	Anti-[electron] neutrino Accept: [electron] anti-neutrino		1		1		
		(ii)	**Lepton number:** 0 = 0 + 1 -1 must be in this order **or** described: lepton number is 0 on LHS. Electron has lepton number of +1, anti-electron neutrino has a lepton number of -1. [1] Accept 4 = 4 + 1 -1 must be in this order. **Charge:** B has one more proton than Be, but appearance of e^- means that charge is conserved **or** +4 = +5 – 1 [+ 0] must be in this order **Or:** 0 $\rightarrow$1 -1 + 0 must be in this order or described **or** $n = p + e^- +$ anti-neutrino so $d = u + e^- +$anti-neutrino so $-\frac{1}{3} = \frac{2}{3} - 1 + 0$ [1]		2		2		
	(c)	(i)	${}^{10}_{4}\text{Be}$ has 6 neutrons **and** ${}^{10}_{5}\text{B}$ has 5 neutrons	1			1		
		(ii)	LHS: udd [1] RHS: uud [1] **N.B.** bod if udd $\rightarrow$ uud + e^- + anti-neutrino seen Award **1 mark** if above seen with quarks assigned to either the e^- and/or anti-neutrino Award **1 mark** for d$\rightarrow$u **or** d in n$\rightarrow$u in p or similar Award **1 mark** if equation written the wrong way around		2		2		
			Question 1 total	**3**	**5**	**0**	**8**	**0**	**0**